Jay Chao

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Education

University of Wisconsin-Madison Dec 2025
Master of Science (M.S.) in Computer Science, GPA: 3.9/4.0 *Madison, Wisconsin*

National Tsing Hua University Jan 2022
Credit Courses in Computer Science *Hsinchu, Taiwan*

- Coursework: Operating Systems, Computer Networks, Data Engineering, Algorithms, HPC, LLMs
- Teaching Assistant: Reinforcement Learning, Mathematics, Database Systems, Software Security

Skills

Languages: Go, Python, C++, Java, C#, SQL, TypeScript/JavaScript
Infrastructure: Kubernetes, KubeRay, Ray, Docker, Argo CD (GitOps), GitHub Actions, Buildkite, Linux
Backend & APIs: gRPC, REST, FastAPI, Spring, .NET, Node.js
Data & Cloud: Apache Airflow, PostgreSQL, MySQL, SQL Server, AWS, Azure
Testing & AI: k6 load testing, unit/e2e testing, OpenAI API

Experience

NeuraSignal – Software Engineer (Contract) Oct 2026 – Present
Ultrasound Simulation Platform – *Python, C++, NumPy*

- Build collision and stochastic Doppler simulators in Python and C++ for a research-grade medical ultrasound simulation platform (4-engineer team), covering signal processing, numerical validation, and automated testing

Open Source – Contributor Jan 2026 – Present
KubeRay & Ray – *Go, Kubernetes, GitHub Actions*

- Traced up to **55** s of extra startup latency per Ray worker pod to a hard-coded init-container resource limit; opened a PR (with unit tests) that derives the limit from the Ray container's resources
- Fixed History Server request failures for suspended RayClusters by checking the controller's Suspended condition instead of the spec flag, eliminating a race with the log collector's final flush
- Reduced manual CI work for maintainers: merged a flaky e2e-test fix and a stale issue/PR bot, and opened a **/retest** workflow that reruns failed GitHub Actions and Buildkite jobs from a PR comment without executing untrusted PR code

Apache Airflow – *Python, TypeScript, React*

- Merged 5 PRs across Airflow's React UI and provider packages, including a UI rendering fix and a migration of connection metadata to declarative YAML
- Wrote the locale skill for Airflow's AI translation agent, defining terminology and style rules that guide LLM output

IBM – Software Engineer Intern Jul 2025 – Aug 2025
Cloud-Native Modernization – *C++, gRPC, Docker, Kubernetes, Argo CD, Linux, k6*

- Built a http bridge for a closed-source legacy CORBA/gRPC service so Kubernetes could health-check it, automating a manual validation workflow and cutting its run time by **50%**
- Cut deployment time from 30 to 5 minutes **(6x)** by replacing manual deploys with GitOps releases on Kubernetes via Argo CD, enabling blue-green deployments
- Built a k6 gRPC load-testing suite that simulates **1,000** concurrent users to pinpoint latency bottlenecks; presented the prototype and findings in a live demo

Projects

Mini ERP: Purchasing & Inventory System – *C#, ASP.NET Core, SQL Server, Azure, OpenAI API* 2026

- Built a procurement system covering request, tiered approval, purchase order, partial receiving, and inventory, with controls that block self-approval and conflicting concurrent edits
- Made the supplier on-time-delivery report **5x faster** (992 ms → 197 ms on 100K requisitions) by tracing the slowdown to a per-row time-zone conversion in a SQL view
- Added natural-language queries with the OpenAI API, read-only by design: generated SQL runs as a database user that can only read approved views